Coherence Relativity I: Coherence Frames, Frame Invariance, and the Born Rule

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We propose that the quantum measurement problem is a frame problem, resolved by recognizing that coherence is relative to a choice of coherence frame—analogous to the relativity of simultaneity in special relativity. We introduce coherence relativity, a geometric framework in which physical reality is described on a joint manifold $\mathcal{M} \times \Sigma$, where $\mathcal{M}$ is emergent spacetime and Σ is a coherence manifold whose points represent distinct observational perspectives. In this framework, wave function collapse is not a dynamical process but a transformation between coherence frames induced by measurement interactions. We present a derivation of the Born rule for pure states—that measurement probabilities equal squared amplitudes—as the unique probability measure invariant under coherence-frame relabelings and splittings, given additivity and continuity. Extension to mixed states via purification is addressed in Paper 2. This derivation unifies insights from Gleason's theorem, Zurek's envariance, and operational reconstruction programs within a single geometric structure. We demonstrate the framework with the double-slit experiment in two coherence frames and identify six experimentally testable signatures distinguishing Coherence Relativity from standard quantum mechanics: non-linear visibility scaling, frame-transformation hysteresis, coherence-space curvature effects, modified gravitational decoherence, measurable transformation timescales, and coherence revival. The most accessible predictions—erasure hysteresis and transformation timescales—are testable with current superconducting qubit and trapped ion platforms.

I. INTRODUCTION

A. The Measurement Problem as a Frame Problem

The quantum measurement problem—how definite outcomes emerge from superposed states—has persisted for nearly a century despite the extraordinary empirical success of quantum mechanics. At its core lies a conceptual asymmetry: the formalism describes systems evolving unitarily, yet observations yield definite results governed by the Born rule, $p_k = |c_k|^2$. The standard resolution appends a measurement postulate that operates outside the unitary framework, creating what Penrose called "the most important open problem in physics" and what Bell termed "the shifty split" between quantum and classical domains [1].

Attempts to resolve this tension have proceeded along several lines. The mathematical framework for this tension was established by von Neumann [2]. Gleason's theorem [3] showed the Born rule follows from noncontextuality, but left its physical origin unexplained. Zurek's envariance [4, 5] derived probabilities from entanglement symmetry, though critics note potential circularity. The Deutsch–Wallace approach [6, 7] obtained Born weights from rationality axioms in an Everettian universe. The decoherence program [8, 9] explained interference suppression while assuming the Born rule. Relational quantum mechanics [10] and QBism [11] reinterpret the formalism without deriving probabilities. Objective collapse models [12–15] modify the dynamics itself.

Each captures an essential aspect. Yet none simultaneously provides: (i) a derivation of the Born rule from physical principles, (ii) a geometric framework explaining *why* those principles hold, (iii) testable predictions, and (iv) ontological clarity about frame-invariant content.

B. Coherence as Relativity

We propose that the measurement problem is not a dynamical problem requiring new physics, but a *frame problem*—analogous to the shift from Newtonian absolute time to the relativity of simultaneity.

The central idea: **coherence is frame-relative**. What appears as a definite outcome in one coherence frame appears as a coherent superposition in another, just as events simultaneous in one inertial frame are nonsimultaneous in another. There is no absolute fact about whether a system "has collapsed."

We formalize this through four elements:

1. **Coherence space** Σ: A manifold whose points represent coherence frames. Physical reality is described on $\mathcal{M} \times \Sigma$.

2. **Frame transformations**: Maps $U_{F \to F'}$ between coherence frames, analogous to Lorentz transformations.

3. **Born rule as frame-invariant measure**: Requiring invariance under coherence relabelings and frame splittings, plus additivity and continuity, uniquely fixes $\mu(\psi_i) = |c_i|^2$.

4. **Tensor field** T_{AB}: A field on $\mathcal{M} \times \Sigma$ encoding the information geometry of coherence frames.

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A coherence frame is not merely a choice of orthonormal basis. A basis change is a passive relabeling within a fixed descriptive structure. A coherence frame, by contrast, specifies which degrees of freedom stably encode definiteness under the system–environment coupling that implements measurement. Formally, a coherence frame is an equivalence class of bases together with a pointer observable selected by dynamical stability. The coherence manifold Σ therefore parameterizes not just representations, but physically distinct descriptive perspectives distinguished by which observables behave classically under interaction. The framework becomes nontrivial because transitions between such frames carry an information-geometric cost measured by the Fubini–Study metric and, in general, exhibit path dependence through the tensor field T_{AB}.

The novelty of this work is not interpretational but structural: the identification of the coherence manifold $\Sigma = U(d)/T^d$ as a geometric arena on which a frame-invariance principle uniquely fixes the Born measure. Collapse, coherence, and classicality then appear as frame-dependent descriptions of a single underlying reality, while probabilities arise as the only quantities invariant under transformations on Σ.

C. Organization

Section II introduces coherence frames and the geometry of Σ. Section III develops frame transformations and invariants. Section IV derives the Born rule. Section V resolves standard quantum paradoxes. Section VI discusses implications, and Sec. VII summarizes.

II. COHERENCE FRAMES AND THE GEOMETRY OF QUANTUM DESCRIPTION

A. Definitions

Definition 1 (Coherence Space)**.** *Let $\mathcal{H}$ be a Hilbert space of dimension d. The* ***coherence space*** *Σ is the space of orthonormal bases of $\mathcal{H}$, modulo physically irrelevant phase redundancies:*

$$\Sigma = U(d)/T^d, \tag{1}$$

where $U(d)$ is the unitary group and T^d is the maximal torus. Mathematically, Σ is the complete flag manifold $\mathrm{Fl}(\mathbb{C}^d)$*, a compact Kähler manifold of real dimension d^2-d [16, 17]. Its Fubini–Study geometry (Sec. II C) provides the natural information-geometric metric [18].*

Definition 2 (Coherence Frame)**.** *A* ***coherence frame*** *$F \in \Sigma$ is a choice of preferred orthonormal basis $\{|i\rangle_F\}$ together with a specification of which degrees of freedom encode definiteness.*

Definition 3 (State Representation)**.** *For physical reality R, the state in frame F is*

$$|\psi\rangle_F = \sum_i c_i^{(F)} |i\rangle_F , \tag{2}$$

where the coefficients $c_i^{(F)}$ depend on the frame. The same R has different representations in different frames: $|\psi\rangle_F (R) \neq |\psi\rangle_{F'} (R)$ in general.

Here $\mathcal{R}$ is defined operationally as the equivalence class of all frame-dependent descriptions $\{|\psi\rangle_F\}_{F\in\Sigma}$ connected by frame transformations, under the equivalence relation $|\psi\rangle_F \sim |\psi\rangle_{F'}$ if and only if there exists $U_{F\to F'}$ such that $|\psi\rangle_{F'} = U_{F\to F'} |\psi\rangle_F$. Crucially, $\mathcal{R}$ carries the *pointer structure* as additional data: the specification of which degrees of freedom encode definiteness under system–environment coupling. This distinguishes $\mathcal{R}$ from a mere ray in Hilbert space—two states related by a unitary that does not preserve the pointer observable belong to different equivalence classes. Formally, $\mathcal{R}$ is a point on $\mathcal{M} \times \Sigma$ rather than in $\mathcal{H}$, and its frame-invariant content consists of the quantities listed in Theorem 1: Born measures, inner products, spectra, and entanglement entropy.

B. Examples

Single qubit. For $d = 2$, Σ is parameterized by the Bloch sphere. The frame F_z corresponds to $\{|0\rangle , |1\rangle\}$; the frame F_x to $\{|+\rangle , |-\rangle\}$. A state $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$ in F_z becomes $|\psi\rangle = \frac{\alpha+\beta}{\sqrt{2}} |+\rangle + \frac{\alpha-\beta}{\sqrt{2}} |-\rangle$ in F_x. Coherence is not a property of the state alone but of the state-in-a-frame.

Double slit. The interference frame F_{int} has basis states that are path superpositions; the which-path frame F_{wp} has path eigenstates. Both describe the same setup (Sec. V).

Bipartite system. For $\mathcal{H}_A \otimes \mathcal{H}_B$, the global frame F_{global} exhibits entanglement. The local frame F_A (tracing out B) shows mixedness instead. Decoherence is reinterpreted as a frame change.

C. Metric Structure

The coherence space inherits the Fubini–Study metric:

$$G_{\lambda\lambda} = \mathrm{Re}[\langle\partial_\lambda\psi|\partial_\lambda\psi\rangle - \langle\partial_\lambda\psi|\psi\rangle\langle\psi|\partial_\lambda\psi\rangle] . \tag{3}$$

This measures the information-geometric distance between nearby frames. The geodesic distance $d(F, F') = \int \sqrt{G_{\lambda\lambda}}\, d\lambda$ gives the shortest path in Σ. The distinguishability parameter $\lambda \equiv \sqrt{1 - |\langle W_L|W_R\rangle|^2}$ follows the standard Englert–Greenberger–Yasin (EGY) complementarity form [19, 20], satisfying $\mathcal{V}^2 + \lambda^2 = 1$ where $\mathcal{V} = |\langle W_L|W_R\rangle|$ is the standard quantum-mechanical visibility. Braunstein and Caves [21] established that the

Fubini–Study metric measures statistical distinguishability as $ds^2 \propto 1-|\langle\psi|\phi\rangle|^2$, making λ a natural information-geometric coordinate along the decoherence trajectory in Σ.

Figure 1 shows $G_{\lambda\lambda}(\lambda)$ and its inverse $d\lambda/ds = 1/\sqrt{G_{\lambda\lambda}}$ computed numerically for a multi-mode decoherence model ($N = 10$ collectively coupled environment modes). Under the EGY parameterization, the metric is finite at $\lambda \to 0$ for all models ($G_{\lambda\lambda}(0) = 1/2$, reflecting the geometric accessibility of the coherent frame) and diverges at $\lambda \to 1$ (rigidity of fully established classical records). For the single-mode analytic case ($N = 1$), $G_{\lambda\lambda} = 1/[2(1-\lambda^2)]$, which is monotone increasing. The $N = 10$ multi-mode model has the same qualitative shape under the EGY parameterization. The inverse $d\lambda/ds = 1/\sqrt{G_{\lambda\lambda}}$ decreases monotonically from $\sqrt{2}$ at $\lambda = 0$ to 0 at $\lambda \to 1$, showing that distinguishability accumulates most rapidly at the coherent endpoint and slows progressively as classical records are established. A coordinate-invariant and model-independent definition of the Markov transition is developed in Paper 2 via the $\delta\lambda$ formalism.

D. The Tensor Field T_{AB}

The metric (3) captures the symmetric, information-geometric cost of moving between nearby frames. We now define the full tensor field T_{AB} on Σ that encodes both metric and connection structure.

Definition 4 (Coherence Tensor Field)**.** *Let x^A denote local coordinates on Σ, with $A, B, \ldots$ indexing coherence-space directions. The* ***coherence tensor field*** *is*

$$T_{AB}(x) = G_{AB}(x) + \Omega_{AB}(x), \tag{4}$$

where $G_{AB} = G_{BA}$ is symmetric and $\Omega_{AB} = -\Omega_{BA}$ is antisymmetric.

Symmetric part. The metric G_{AB} is the pullback of the Fubini–Study metric onto Σ along the state map $x \mapsto |\psi(x)\rangle$:

$$G_{AB}(x) = \mathrm{Re}[\langle\partial_A\psi|\partial_B\psi\rangle - \langle\partial_A\psi|\psi\rangle\langle\psi|\partial_B\psi\rangle]\,, \tag{5}$$

where $\partial_A := \partial/\partial x^A$. This is positive-semidefinite and measures the information-geometric cost of an infinitesimal frame change δx^A. Equation (3) is the one-dimensional restriction $G_{\lambda\lambda} = G_{AB}\,\delta^A_\lambda\,\delta^B_\lambda$ for the double-slit which-path parameter.

Antisymmetric part. The two-form Ω_{AB} derives from a connection A_A on coherence space, with curvature $F_{AB} = \partial_A A_B - \partial_B A_A$. It encodes non-integrable aspects of frame changes: how coherence frames twist relative to each other along closed loops in Σ. In quasi-equilibrium situations (reversible frame changes), $\Omega_{AB} = 0$ and T_{AB} reduces to the metric. For irreversible processes—such as partial erasure followed by attempted revival—$\Omega_{AB} \neq 0$ generates path-dependent (hysteretic) behavior.

We emphasize that Ω_{AB} is *not* the Berry curvature of the state. Berry curvature characterizes holonomy of the state vector under adiabatic parameter variation and lives on the external parameter manifold. Ω_{AB} is instead the curvature of the *frame bundle* over Σ: it encodes how the pointer structure twists when coherence frames are transported along closed loops in Σ. It vanishes for reversible (unitary) frame changes and is nonzero precisely when a closed loop in Σ fails to return the pointer observable to its initial configuration. The symmetric part G_{AB} reproduces the Fubini–Study metric—the Kähler metric on $\mathbb{CP}^{d-1}$—while Ω_{AB} encodes dynamical irreversibility in the frame bundle, a structure absent from the standard Kähler form. The full $T_{AB} = G_{AB} + \Omega_{AB}$ therefore carries genuinely new geometric content beyond standard projective Hilbert space structure.

Double-slit reduction. For the one-dimensional coherence manifold $\Sigma \simeq [0, 1]$ with which-path parameter λ, the tensor field reduces to the single scalar component $T_{\lambda\lambda} = G_{\lambda\lambda}(\lambda)$ (since $\Omega_{\lambda\lambda} = 0$ trivially in one dimension). The full tensor structure becomes physically relevant for systems with $\dim\Sigma \geq 2$, where closed loops in coherence space are possible.

The complete development of T_{AB}—including the decomposition into $\mathcal{M}$ and Σ coordinates, the cross-term $T_{M\Sigma}$ encoding $\mathcal{M}$-Σ coupling, the $\delta\lambda$ formalism separating frame-lag from decoherence progress, equations of motion, and the connection to emergent spacetime—is the subject of Paper 2. Here we require only the decomposition (4), the metric definition (5), and the qualitative role of Ω_{AB} in generating hysteresis.

E. Analogy with Special Relativity

The coherence-frame structure parallels the inertial-frame structure of special relativity (Table I). The key structural parallel: in SR, there is no absolute simultaneity—it depends on the inertial frame. In Coherence Relativity, there is no absolute fact about whether a system is "in superposition" or "has collapsed"—it depends on the coherence frame.

TABLE I. Structural analogy between special relativity and coherence relativity.

Special Relativity	Coherence Relativity
Spacetime event	Physical reality R
Inertial frame	Coherence frame F
Coordinates $(t, \mathbf{x})$	State representation $\|\psi\rangle_F$
Lorentz transformation	Frame transformation $U_{F\to F'}$
Invariant interval ds^2	Born measure $\mu = \|c_i\|^2$
Minkowski metric	Fubini–Study metric on Σ
Simultaneity (relative)	Coherence (relative)

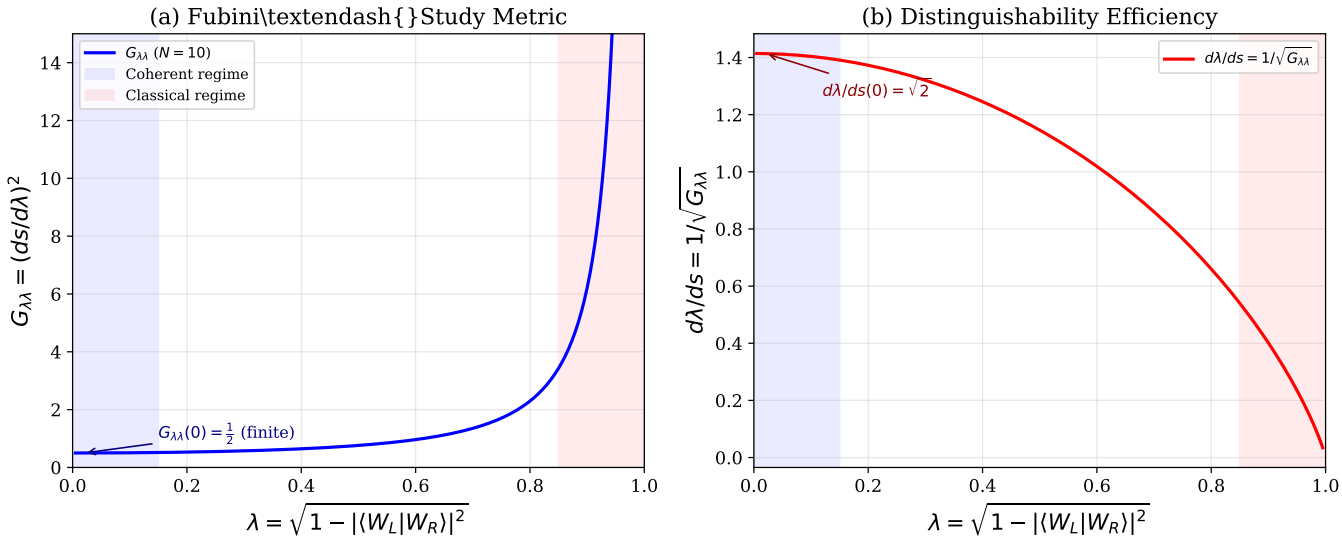

FIG. 1. Geometric structure of the decoherence trajectory under the EGY parameterization $\lambda = \sqrt{1 - |\langle W_L|W_R\rangle|^2}$ (multi-mode model, $N = 10$). **(a)** The Fubini–Study metric $G_{\lambda\lambda} = (ds/d\lambda)^2$: finite at $\lambda \to 0$ (all models: $G_{\lambda\lambda}(0) = 1/2$, geometric accessibility of the coherent frame) and diverging at $\lambda \to 1$ (rigidity of classical records). **(b)** The inverse $d\lambda/ds = 1/\sqrt{G_{\lambda\lambda}}$ starts at $\sqrt{2}$ and decreases monotonically: distinguishability accumulates efficiently near the coherent endpoint and slows as classical records solidify.

III. FRAME TRANSFORMATIONS AND INVARIANTS

A. Frame Transformation Maps

Definition 5 (Frame Transformation)**.** *A **frame transformation** $U_{F\to F'}$ maps state representations: $|\psi\rangle_{F'} = U_{F\to F'}|\psi\rangle_F$. In the Hilbert space picture, $U_{F\to F'}$ is unitary:*

$$|i\rangle_{F'} = \sum_j (U_{F\to F'})_{ji}\, |j\rangle_F \,. \tag{6}$$

Definition 6 (Coherence Relabeling)**.** *A **coherence relabeling** permutes and phase-rotates basis vectors:*

$$U_{F\to F'} : |i\rangle_F \to e^{i\theta_i}\, |\pi(i)\rangle_{F'} \,, \tag{7}$$

where π is a permutation and $\theta_i \in \mathbb{R}$.

B. Group Structure

Frame transformations form a **groupoid** over Σ:

1. Composition: $U_{F\to F''} = U_{F'\to F''} \circ U_{F\to F'}$
2. Identity: $U_{F\to F} = \mathbb{I}$
3. Inverse: $(U_{F\to F'})^{-1} = U_{F'\to F}$

For finite-dimensional $\mathcal{H}$, each $U_{F\to F'}$ has a unitary matrix representation preserving inner products: $\langle\psi|\phi\rangle_F = \langle U\psi|U\phi\rangle_{F'}$.

C. Frame Splitting

Definition 7 (Frame Splitting)**.** *A **frame splitting** transforms a single basis state into an equal-amplitude superposition:*

$$|i\rangle_F \to \frac{1}{\sqrt{m_i}} \sum_{k=1}^{m_i} |i,k\rangle_{F'} \,, \tag{8}$$

where $\{|i,k\rangle_{F'}\}$ are orthonormal in the enlarged space.

Proposition 1 (Existence of Splittings)**.** *For any state $|\psi\rangle_F = \sum_i c_i |i\rangle_F$ with rational $|c_i|^2 = m_i/M$ (where $M = \sum_i m_i$), there exists a splitting $F \to F'$ such that $|\psi\rangle_{F'} = M^{-1/2}\sum_{i,k}|i,k\rangle_{F'}$ is equal-amplitude.*

Proof. Introduce an ancilla Hilbert space $\mathcal{H}_{\text{anc}}$ of dimension M with orthonormal basis $\{|e_j\rangle\}_{j=1}^M$. Partition the ancilla basis into blocks: for each i, assign m_i ancilla states $|e_j\rangle$ with $j \in B_i$, where $|B_i| = m_i$ and the blocks are disjoint with $\bigcup_i B_i = \{1,\dots,M\}$.

Define an isometry $V : \mathcal{H} \to \mathcal{H}\otimes\mathcal{H}_{\text{anc}}$ by

$$V|i\rangle_F = \frac{1}{\sqrt{m_i}} \sum_{j\in B_i} |i\rangle_F \otimes |e_j\rangle \,. \tag{9}$$

Then V preserves inner products: $\langle i|V^\dagger V|i'\rangle = \delta_{ii'}$. Applying V to $|\psi\rangle_F$:

$$V|\psi\rangle_F = \sum_i c_i \cdot \frac{1}{\sqrt{m_i}} \sum_{j\in B_i} |i,e_j\rangle = \frac{1}{\sqrt{M}} \sum_i \sum_{j\in B_i} |i,e_j\rangle \,, \tag{10}$$

where the last equality uses $c_i = \sqrt{m_i/M}$ (up to phases absorbed by relabeling). Identifying $|i,k\rangle_{F'} \equiv |i,e_j\rangle$ for $j \in B_i$ gives the equal-amplitude state $|\psi\rangle_{F'} = M^{-1/2}\sum_{i,k}|i,k\rangle_{F'}$ in the enlarged frame F'. □

Physically, this construction corresponds to refining a measurement by coupling to an ancilla—a standard procedure in quantum information.

The frame splitting embeds Σ into the larger coherence space $\Sigma' = U(dM)/T^{dM}$; Axiom (A1) requires only that the probability measure μ is preserved under this embedding. We do not claim T_{AB} is defined on Σ'; the splitting is used solely to establish the probability assignment via Steps 2–3 of the Born rule derivation, after which the isometry returns to the original Σ. Compatibility of T_{AB} across dimension-changing embeddings is addressed in Paper 2.

D. Frame Invariants

Theorem 1 (Frame Invariants)**.** *The following are invariant under all frame transformations:*

1. *Total probability:* $\sum_i \mu_F(i) = 1$
2. *Born measures of alternatives:* $\mu_F(i) = \mu_{F'}(\pi(i))$
3. *Inner products:* $|\langle\psi|\phi\rangle|^2$ *is frame-independent*
4. *Eigenvalue spectra of observables*
5. *Entanglement entropy* $S(\rho_A)$ *for bipartite systems*

What is *not* invariant: which observables have definite values, whether a state appears coherent or decohered, and the form of the density matrix.

IV. THE BORN RULE AS FRAME-INVARIANT MEASURE

The derivation that follows is Gleason-like in structure but replaces noncontextuality with frame-invariance: the symmetry is not "the same observable measured by different projectors yields the same result" but rather "the same physical situation described in different coherence frames yields the same probabilities."

A. Statement

The key symmetry underlying the derivation is invariance under refinement of alternatives. If two descriptions differ only by coupling to an ancilla that refines a single alternative into multiple orthogonal sub-alternatives of equal amplitude, then no physical probability assignment can distinguish between them. Such refinements are routine in quantum information and correspond operationally to increasing measurement resolution without altering the underlying physical situation. A probability measure that depended on whether or not such a refinement had been performed would be ill-defined, as it would assign different probabilities to physically equivalent descriptions. We formalize this requirement as invariance under frame splitting.

The ancilla splitting construction in Proposition 1 follows the fine-graining technique used in Zurek's envariance derivation [5] and the Deutsch–Wallace approach [6, 7]. The novelty of the present derivation lies not in the construction itself but in its role: rather than establishing envariance or decision-theoretic equivalence, the splitting grounds *frame-invariance on* Σ as the operative symmetry—a geometric condition that holds for $\dim\mathcal{H} \geq 2$ and requires no physical environment or branching ontology.

Theorem 2 (Born Rule from Frame Invariance)**.** *Let* Σ *be the coherence space for* $\mathcal{H}$ *with* $\dim \geq 2$*. Let* μ *be a measure on the alternatives* $\{|i\rangle_F\}$ *satisfying:*

(A1): ***Frame invariance****:* μ *is invariant under coherence relabelings* (7) *and frame splittings* (8)*.*

(A2): ***Additivity****:* μ *is additive on mutually orthogonal alternatives.*

(A3): ***Frame locality:*** $\mu_F(i)$ *depends only on* $|\psi\rangle_F$ *and* $|i\rangle_F$*—not on the history of frame transformations leading to* F*, nor on probability assignments in other frames.*

This is a locality assumption on Σ*, structurally analogous to local causality in special relativity—a postulate reflecting the physical structure of coherence space, not a derived result. It excludes contextual hidden-variable models in which the probability of outcome* i *depends on which other measurements are simultaneously possible. We regard this exclusion as physically motivated: frame-history dependence would prevent coherence frames from constituting a well-defined descriptive language independent of experimental arrangement.*

(A4): ***Continuity****:* $\mu_F(i)$ *is continuous in the coefficients* c_i*.*

Then $\mu_F(i) = |c_i|^2$*, where* $|\psi\rangle_F = \sum_i c_i |i\rangle_F$*.*

B. Derivation

Step 1: Phase independence. By (A1), μ is invariant under coherence relabelings (7), including arbitrary phase rotations θ_i. Hence $\mu_F(i)$ depends on c_i only through $|c_i|$.

Step 2: Equal amplitudes. For $|\psi\rangle_F = N^{-1/2}\sum_{k=1}^{N}|k\rangle_F$, permutation invariance gives $\mu_F(k) = 1/N = |c_k|^2$ for all k.

Step 3: Rational amplitudes. For $|\psi\rangle_F = \sqrt{m/(m+n)}\,|0\rangle_F + \sqrt{n/(m+n)}\,|1\rangle_F$, construct a frame

splitting (Prop. 1) producing an equal-amplitude state of $m+n$ terms. By Step 2 and additivity (A2):

$$\mu_F(0) = \frac{m}{m+n} = |c_0|^2, \quad \mu_F(1) = \frac{n}{m+n} = |c_1|^2. \quad (11)$$

Step 4: General amplitudes. Approximate $|c_i|^2$ by rationals $r_i^{(n)}$ with $|r_i^{(n)} - |c_i|^2| < 1/n$. By Step 3, $\mu^{(n)}(i) = r_i^{(n)}$. By continuity (A4), $\mu_F(i) = \lim_{n\to\infty} \mu^{(n)}(i) = |c_i|^2$. □

C. Comparison with Previous Derivations

Table II summarizes the relationship to prior work. The structural parallel with Zurek's envariance [5] warrants explicit comparison. Both derivations use a splitting step to reduce general amplitudes to equal amplitudes. However, they differ in three respects: (i) Zurek's envariance requires a physical environment and invokes swap symmetry of the system–environment state, which critics [4] have argued presupposes the Born rule; our splitting (Prop. 1) is a mathematical theorem about Hilbert space embeddings, requiring no environment. (ii) Envariance acts on a bipartite state $|\psi\rangle_{SE}$; frame invariance acts on the geometric structure of Σ itself, making it independent of any particular physical setup.

A precise comparison with Gleason's theorem is warranted. Gleason's non-contextuality (NC) is a *global* condition: a projector P must yield the same probability regardless of which resolution of the identity it belongs to—across *all* POVMs simultaneously. Axiom (A1) requires something strictly weaker: that isometric embeddings of the form in Proposition 1—equal-amplitude frame splittings via ancilla coupling—preserve probability assignments. This is a *local geometric* condition on Σ, placing no constraint on probabilities for projectors in unrelated measurement contexts. The logical relationship is: NC $\Rightarrow$ (A1), but (A1) $\not\Rightarrow$ NC. We note additionally that Gleason's theorem requires $\dim \mathcal{H} \geq 3$; the present derivation holds for $\dim \geq 2$ because the splitting argument operates on the ancilla-extended space, bypassing the qubit obstruction.

(iii) The coherence-relativity derivation makes the *geometric origin* of the symmetry manifest: frame invariance is the requirement that physics does not depend on which coherence frame is used for description, just as Lorentz invariance requires independence of inertial frame.

The present derivation establishes $\mu_F(i) = |c_i|^2$ for pure states via (A1)–(A4). Extension to mixed states follows by purification: any mixed state ρ on $\mathcal{H}_S$ is the partial trace of a pure state on $\mathcal{H}_S \otimes \mathcal{H}_R$, and the pure-state result recovers $\mathrm{Tr}(\rho|i\rangle\langle i|)$ by tracing out $\mathcal{H}_R$. The circularity objection applying to envariance—which requires a physical environment and swap symmetry—does not apply here: frame splittings (Prop. 1) are mathematical theorems about Hilbert space embeddings requiring no environment.

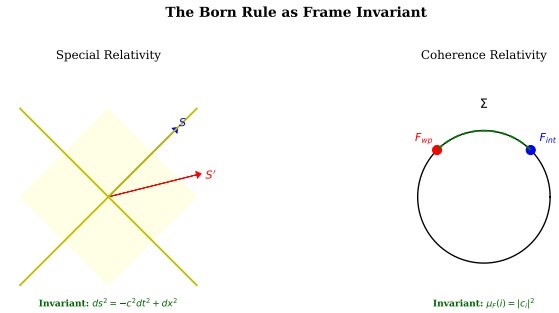

FIG. 2. The Born rule as frame invariant. Left: in special relativity, the interval ds^2 is invariant under Lorentz transformations. Right: in Coherence Relativity, the Born measure $\mu_F(i) = |c_i|^2$ is invariant under coherence-frame transformations on Σ.

D. Physical Interpretation

The Born rule is not a postulate, a frequency statement, or a consequence of branching. It is the **unique frame-invariant probability assignment** on Σ—the quantum analog of the invariant interval ds^2 in special relativity (Fig. 2).

V. PARADOX RESOLUTIONS AND WORKED EXAMPLES

A. Double-Slit Experiment

Consider the double-slit with two frames (Fig. 3):

- F_{int}: $|\psi\rangle_{F_{\text{int}}} = \frac{1}{\sqrt{2}}(|L\rangle + |R\rangle)$. Interference visible.
- F_{wp}: $|\psi\rangle_{F_{\text{wp}}} = \frac{1}{\sqrt{2}}(|L\rangle\,|W_L\rangle + |R\rangle\,|W_R\rangle)$. Paths definite.

The transformation $U_{F_{\text{int}}\to F_{\text{wp}}}$ is realized by detector coupling. Frame invariants: $\mu(L) = \mu(R) = 1/2$ in both frames. Frame-dependent: visibility of interference fringes, off-diagonal density matrix elements.

B. Wigner's Friend

Wigner's friend [23] performs a measurement inside a laboratory (frame F_{friend}: state has collapsed to $|\uparrow\rangle$ or $|\downarrow\rangle$). Wigner, outside the lab, describes the joint system in F_{Wigner}: the friend–system composite remains in superposition. In Coherence Relativity, both descriptions are correct *in their respective frames*. There is no contradiction, just as there is no contradiction between observers disagreeing about simultaneity in SR.

TABLE II. Comparison of Born rule derivations.

Derivation	Symmetry Used	Physical Content	Assumes Born?	Result
Gleason (1957)	Non-contextuality	Mathematical ($\mathcal{H}$)	No	$\mu = \mathrm{Tr}(\rho P)$
Zurek (2005)	Envariance	Entanglement + environment	Debated	$\mu = \lvert c_i \rvert^2$
Deutsch–Wallace	Rational equivalence	Many-worlds ontology	Debated	$\mu = \lvert c_i \rvert^2$
Masanes et al. [22]	Operational axioms	Compositional structure	No	Born rule unique
This work	**Frame invariance**	**Geometric (Σ, T_{AB})**	**No**	$\boldsymbol{\mu = \lvert c_i \rvert^2}$

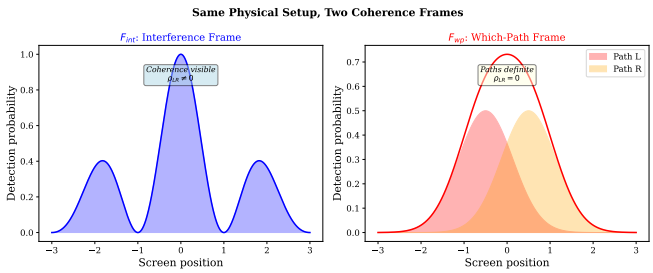

FIG. 3. Double-slit experiment in two coherence frames. Left: interference frame F_{int} shows the characteristic fringe pattern. Right: which-path frame F_{wp} resolves individual paths with no interference. The Born measure $\mu(L) = \mu(R) = 1/2$ is invariant.

C. EPR and Bell Nonlocality

Einstein, Podolsky, and Rosen [24] argued that quantum mechanics is incomplete based on correlations between separated systems. For an entangled pair in the singlet state, Alice's and Bob's local frames each see mixed states. The global frame sees entanglement. Bell inequality violation is a frame-invariant fact (it depends only on $|\langle\psi|\phi\rangle|^2$ terms). The correlations are not "caused" by collapse but are geometric properties of the entangled state visible from any frame.

D. Schrödinger's Cat

The cat-in-a-box scenario involves two frames: F_{outside} (cat in superposition) and F_{inside} (cat alive or dead). Opening the box is a frame transformation $U_{F_{\mathrm{outside}} \to F_{\mathrm{inside}}}$, not a physical collapse. The cat's definite state exists in F_{inside} at all times; it simply isn't accessible from F_{outside} without the frame transformation.

VI. DISCUSSION

A. Resolution of the Measurement Problem

Coherence Relativity offers a relational resolution of the measurement problem: "collapse" is a frame transformation, not a dynamical process. The Born rule is not postulated but derived as the unique frame-invariant measure. Definite outcomes exist in appropriate coherence frames; superposition exists in others. There is no need to explain *when* or *how* collapse occurs, because collapse is not a physical event—it is a change of description. Crucially, no frame ever loses coherence: what changes is the geometric relationship between alternatives, as distinguishability becomes redundantly encoded in the environment. Coherence is not destroyed but *re-described* from the perspective of a different frame.

B. Quasipotential Landscape

The tensor field T_{AB} (Sec. II D) generates a quasipotential landscape on Σ via an action principle. For a path $\gamma : s \mapsto x(s)$ through coherence space with effective drift $F^A(x)$ (driven by system–environment coupling) and noise scale D, define the action

$$\mathcal{S}[\gamma] = \frac{1}{4D} \int_0^1 (\dot{x}^A - F^A)\, G_{AB}\, (\dot{x}^B - F^B)\, ds. \qquad (12)$$

This is the Freidlin–Wentzell large-deviation functional [25], which defines the minimum-cost path for stochastic excursions away from an attractor. The quasipotential is $V(x) = \inf_\gamma \mathcal{S}[\gamma]$, where the infimum is over paths from a reference frame to x (Fig. 4). Minima of V correspond to metastable classical frames (pointer states), saddle points determine transition rates, and the stationary distribution concentrates as $p(x) \propto \exp(-V(x)/\epsilon)$ for small effective noise ϵ. This provides a geometric explanation for einselection: environment-induced superselection is the dynamical relaxation to V-minima. In the purely geometric limit (zero drift, zero noise), the quasipotential reduces to the Fubini–Study distance to the classical frame, $d_\Sigma(\lambda) = \int_\lambda^1 \sqrt{G_{\lambda\lambda}}\, d\lambda'$, which we plot in Figs. 4 and 5.

The novel application is the identification of Σ as the arena and G_{AB} as the cost tensor—a construction with no prior precedent in the quantum decoherence literature. We conjecture that F^A, when derived explicitly from the system–environment interaction Hamiltonian H_{SE}, drives coherence-space trajectories to einselected pointer-state frames; this is consistent with Zurek's einselection picture [4] and is the primary object of Paper 2. Here F^A enters as a phenomenological drift term whose form is constrained by the requirement that V-minima reproduce known pointer states in tractable models.

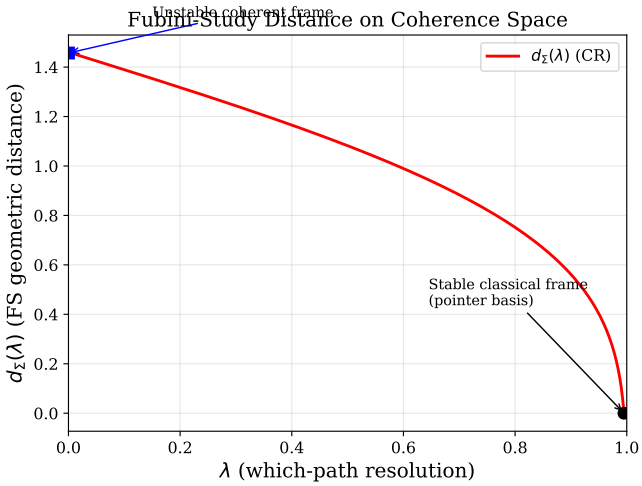

FIG. 4. Geometric distance $d_\Sigma(\lambda) = \int_\lambda^1 \sqrt{G_{\lambda\lambda}}\, d\lambda'$ on coherence space (EGY parameterization, $N = 10$), measuring the Fubini–Study distance from λ to the classical frame. In the zero-noise limit this coincides with the quasipotential V of Eq. (12). The finite starting value $d_\Sigma(0) \approx 1.46$ reflects the geometric accessibility of the coherent frame. The steep drop near $\lambda \to 1$ reflects the divergent metric (rigidity of classical records). The global minimum at $\lambda = 1$ (pointer basis) represents the stable classical frame.

The accumulated Fubini–Study length $s(\lambda) = \int_0^\lambda \sqrt{G_{\lambda\lambda}}\, d\lambda'$ quantifies the total geometric distance traversed from the coherent frame. For the multi-mode model ($N = 10$), the total length is $s(1) \approx 1.46$; for $N = 1$, $s(1) = \pi/(2\sqrt{2}) \approx 1.11$. Under the EGY parameterization, $s(\lambda)$ grows from 0 at $\lambda = 0$ and accelerates near $\lambda \to 1$ (where $G_{\lambda\lambda}$ diverges). The quasipotential $V(\lambda) = s(1) - s(\lambda)$ therefore decreases steeply near $\lambda \to 1$, reflecting the increasing geometric cost of maintaining classical records. This is structurally analogous to a potential landscape where the classical fixed point at $\lambda = 1$ sits at the bottom of a well with walls that steepen approaching the pointer basis—the system must traverse the full geometric length $s(1)$ to fully establish classical distinguishability.

C. Initiation State and Coherence-to-Pointer Transition

A minimal initiation model is given by an initially coherent system–environment state

$$|\Psi_0\rangle = \sum_i c_i\, |i\rangle_S \otimes |E_0\rangle, \tag{13}$$

with interaction map

$$U_{SE}:\ |i\rangle_S|E_0\rangle \mapsto |i\rangle_S|E_i\rangle. \tag{14}$$

Define the decoherence distinguishability coordinate

$$\lambda_D := 1 - \big|\langle E_i|E_j\rangle\big|, \tag{15}$$

with $\lambda_D \in [0, 1]$. In this paper, the geometric order parameter λ is taken to be a monotone calibration of λ_D; unless otherwise stated, we work in the gauge $\lambda \equiv \lambda_D$. Then $\lambda_D = 0$ corresponds to fully coherent alternatives and $\lambda_D \to 1$ to pointer-resolved records. The reduced state

$$\rho_S(\lambda_D) = \mathrm{Tr}_E[|\Psi(\lambda_D)\rangle\langle\Psi(\lambda_D)|] \tag{16}$$

approaches diagonality in a dynamically stable pointer basis when decohering couplings dominate recohering channels. In this framework, classical spacetime is not assumed at $\lambda_D = 0$; rather, it is an effective description associated with the high-distinguishability sector ($\lambda_D \to 1$) and the induced macroscopic observables. C^1 regularity is adopted here as a transition-level condition on the coherence-to-classical map: the trajectory on Σ is smooth in information-geometric distance, and the induced macroscopic fields are continuous with continuous first derivatives across the crossover. For the light-cone profile $A = z$, this regularity is satisfied; for warped profiles $A(r, z)$, the full embedding-level check is deferred to Paper 2, without requiring a brane embedding in the present section.

D. Quantum Darwinism and Objectivity

Coherence Relativity naturally accommodates quantum Darwinism [26]: when multiple environment fragments independently acquire the same information about the system, they select a preferred coherence frame. In Coherence Relativity language, the redundant encoding of pointer-state information across environment fragments [27, 28] is consistent with the narrowing of the quasipotential well around the classical frame as the number of monitoring fragments N increases: each additional fragment independently constrains the preferred coherence frame, geometrically sharpening the V-minimum in a manner made precise in Paper 2. The present paper notes only the structural consistency between the QD redundancy criterion and the geometric quasipotential picture; a quantitative derivation requires the explicit form of F^A from H_{SE}.

E. Testable Predictions

Before stating the predictions, we identify the physical origin of the deviations from standard quantum mechanics. The predictions do *not* arise from modified Hamiltonian dynamics, alterations to the unitary group, or replacement of the Born rule—the full unitary structure of quantum mechanics is preserved on $\mathcal{H}$. The deviations arise instead from the *geometric structure of* Σ: specifically, the Fubini–Study metric $G_{\lambda\lambda}$ and the quasipotential $V(\lambda)$, which standard quantum mechanics does not model. Standard QM implicitly treats all traversals of

the decoherence parameter λ as metrically equivalent; Coherence Relativity predicts that physically traversed paths are weighted by $G_{\lambda\lambda}$. We acknowledge that the geometric structure of Σ constitutes additional structure beyond standard QM's formalism; this framework is therefore a minimal *geometric extension* that becomes empirically distinguishable from standard QM precisely through predictions 1–6 below.

Predictions (1) (non-linear visibility) and (5) (transformation timescale) follow directly from $G_{\lambda\lambda}$ via the geodesic integral (Eq. 18); they require no dynamical assumptions beyond the metric. Predictions (2) (hysteresis $\delta\mathcal{V}$), (3) (curvature scaling), (4) (gravitational decoherence modification), and (6) (coherence revival) additionally invoke the stochastic action $\mathcal{S}[\gamma]$ and drift F^A, whose explicit derivation from system–environment coupling is deferred to Paper 2. The order-of-magnitude estimates for these four predictions are motivated by the geometric structure but should be understood as indicative rather than derived. The quantitative falsifiability of predictions (1) and (5) is therefore independent of Paper 2.

Six experimentally testable signatures distinguish Coherence Relativity from standard QM:

1. **Non-linear visibility**: $\mathcal{V}(\lambda)$ deviates from $\sqrt{1-\lambda^2}$ due to $G_{\lambda\lambda}$ curvature (Fig. 5).

2. **Frame-transformation hysteresis**: Erasure paths $\lambda : 0 \to 1 \to 0$ show path-dependent visibility recovery.

3. **Coherence-space curvature**: Many-body systems exhibit $G_{\lambda\lambda}$ scaling deviations from mean-field predictions.

4. **Modified gravitational decoherence**: Gravitational contributions to T_{AB} produce mass-dependent frame-transformation rates differing from Diósi–Penrose [14, 15].

5. **Transformation timescales**: Weak measurements probe finite frame-transformation duration, resolvable with current qubit platforms.

6. **Coherence revival**: Closed systems show partial coherence recovery forbidden in irreversible-collapse models [12, 29].

1. *Quantitative estimates*

We provide order-of-magnitude estimates for the two most accessible predictions.

Frame-transformation hysteresis. Consider a qubit coupled dispersively to a readout resonator (dispersive shift $\chi/2\pi \sim 1$ MHz). Prepare the state $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, drive the which-path parameter through a cycle $\lambda : 0 \to \lambda_{\max} \to 0$ by pulsing the readout, then measure visibility. In standard QM with a closed erasure cycle,

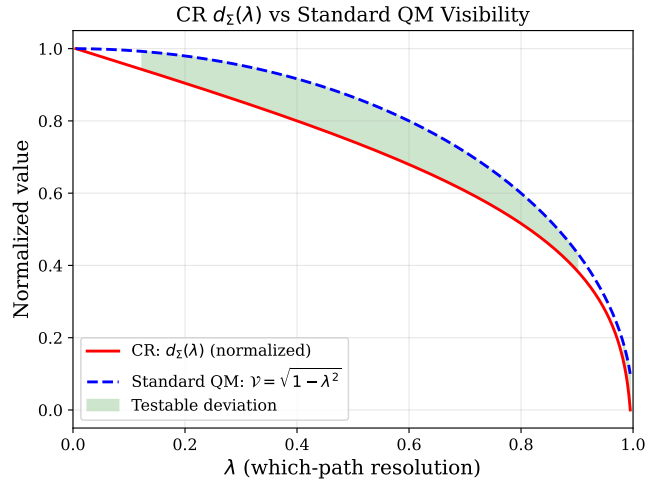

FIG. 5. Comparison of the normalized Coherence Relativity geometric distance $d_\Sigma(\lambda)/d_\Sigma(0)$ (solid, red) with standard QM visibility $\mathcal{V} = \sqrt{1-\lambda^2}$ (dashed, blue) under the EGY parameterization $\lambda = \sqrt{1 - |\langle W_L|W_R\rangle|^2}$. The CR curve deviates from the standard QM prediction, reaching a maximum difference $\Delta \approx 0.12$ at $\lambda \approx 0.52$.

$\mathcal{V}_{\text{revival}} = \mathcal{V}_{\text{initial}}$. In Coherence Relativity, the action cost of the cycle is

$$\Delta\mathcal{S} = \frac{1}{4D} \oint (\dot{\lambda} - F^\lambda)^2 G_{\lambda\lambda}\, ds > 0, \tag{17}$$

A dimensional estimate, connecting $D \sim 1/T_2$ and using the Ricci scalar $\mathcal{R}_\Sigma = 8$ of the qubit coherence space (CP^1 with Fubini–Study metric radius $r = 1/2$, giving $\mathcal{R} = 2/r^2 = 8$ [16]), yields a visibility deficit $\delta\mathcal{V} \sim (\tau_{\text{frame}}/T_2)^2 \cdot \mathcal{R}_\Sigma$, where $\tau_{\text{frame}} \sim 1/\chi \approx 1\ \mu$s and $T_2 \approx 70\ \mu$s. This gives $\delta\mathcal{V} \sim 2 \times 10^{-3}$. A rigorous derivation connecting $\Delta\mathcal{S}$ to the visibility deficit via a specific noise model is deferred to Paper 2; the estimate here serves as an order-of-magnitude guide to experimental sensitivity requirements.

We stress that this hysteresis does not require unitarity violation. The full system–environment state evolves unitarily throughout; the visibility deficit $\delta\mathcal{V}$ is a property of the *reduced* dynamics on Σ—the action cost $\Delta\mathcal{S} > 0$ (Eq. 17) accumulated along the erasure–revival cycle under the effective stochastic drift F^λ. In standard QM the path through Σ carries no metric weight, so $\mathcal{V}_{\text{revival}} = \mathcal{V}_{\text{initial}}$. In Coherence Relativity the Fubini–Study metric renders forward and reverse paths geometrically inequivalent, producing the predicted deficit. The distinction is analogous to reversible vs. irreversible thermodynamic paths: global energy conservation holds in both, but accessible work differs.

Transformation timescales. The geodesic distance in Σ between the coherent frame ($\lambda = 0$) and the pointer frame ($\lambda = 1$) sets a minimum transformation duration:

$$\tau_{\text{frame}} = \frac{1}{v_{\max}} \int_0^1 \sqrt{G_{\lambda\lambda}(\lambda)}\, d\lambda, \tag{18}$$

where $v_{\max}$ is bounded by the measurement coupling strength. For a transmon qubit with $\chi/2\pi = 1$ MHz coupled to a single readout resonator ($N = 1$), the metric takes the form $G_{\lambda\lambda} = 1/[2(1-\lambda^2)]$ under the EGY parameterization, and the geodesic integral evaluates analytically to $\int_0^1 \sqrt{G_{\lambda\lambda}}\, d\lambda = \pi/(2\sqrt{2}) \approx 1.1$, giving $\tau_{\text{frame}} \gtrsim 1.1\ \mu$s. For multi-mode environments ($N = 10$, Fig. 1), the integral over the experimentally accessible range evaluates to ≈ 1.5. Both timescales are resolvable via weak-measurement trajectories [29]: monitoring the qubit with a weak probe during the frame transformation should reveal a finite-duration transition rather than an instantaneous jump, with the transition profile reflecting the monotone $G_{\lambda\lambda}(\lambda)$ from Fig. 1. Since the physical decoherence rate satisfies $ds/dt = \sqrt{G_{\lambda\lambda}}\, d\lambda/dt$, the metric directly shapes the temporal trajectory: rapid state evolution near the coherent endpoint (where $G_{\lambda\lambda} \approx 1/2$ is small) and progressively slower passage as classical records are established (where $G_{\lambda\lambda}$ grows large).

Remark on universality. The timescales above are *not* universal predictions of the framework: they depend on the coupling strength $v_{\max}$ that sets the maximum rate of motion through Σ. What *is* universal is the geometric constraint (18)—a minimum geodesic distance must be traversed—and the characteristic shape of $G_{\lambda\lambda}(\lambda)$. Different experimental platforms (femtosecond photonic vs. microsecond superconducting) will exhibit different absolute timescales because their couplings differ; the testable content is the *ratio* of transformation duration to coupling strength and the path-dependent structure of the metric.

F. Relationship to Existing Frameworks

Coherence Relativity is closest in spirit to relational quantum mechanics [10] but differs in three respects: (i) Σ has concrete geometric structure; (ii) the Born rule is derived, not assumed; (iii) the tensor field T_{AB} provides dynamics absent in RQM.

Compared to QBism [11], Coherence Relativity is realist: frame-invariant content (Born measures, inner products, spectra) constitutes observer-independent physical fact.

Compared to many-worlds [30, 31], Coherence Relativity posits one reality and many frames, not many worlds and one agent. The logical structure of the Born rule derivation parallels the Deutsch–Wallace approach, but replaces branching ontology with geometric frame-invariance.

The quantum reference frame (QRF) program [32–34] shares Coherence Relativity's emphasis on frame-dependence: Giacomini et al. showed that entanglement and superposition are QRF-dependent [33], and perspective-neutral approaches [35, 36] seek frame-invariant structures. Coherence Relativity differs in scope: QRFs transform between quantum systems serving as references, while coherence frames transform between *descriptive bases* for a fixed system. The coherence space Σ provides a global geometric arena absent in QRF treatments, and the Born rule emerges as its unique invariant measure rather than being assumed. Recent work showing that the sum of entanglement and subsystem coherence is invariant under QRF transformations [37] provides independent support for the principle that frame-invariant quantities—not frame-dependent descriptions—constitute physical reality. Figure 6 summarizes the structural distinction.

G. Geometric Emergence Programs

Coherence relativity belongs to a broader contemporary movement in which geometric objects are more fundamental than the physical principles they encode. The Amplituhedron program of Arkani-Hamed and Trnka [38] encodes scattering amplitudes in N=4 Super Yang-Mills theory as the volume of a purely geometric object in Grassmannian space—with no Feynman diagrams, no spacetime, and without assuming locality or unitarity. These emerge from the geometry. The structural parallel to Coherence Relativity is direct: just as the Born rule emerges here as the unique invariant measure on Σ rather than being postulated, unitarity emerges from the Amplituhedron rather than being assumed. Both programs suggest that the deepest level of physical description is geometric, and that the principles physicists have long taken as foundational are in fact consequences of geometric structure. The connection between the coherence manifold Σ and the positive Grassmannian underlying the Amplituhedron is an open question warranting future investigation.

Minimal ontology. To summarize the interpretive commitments: Coherence Relativity posits *one reality* R, *many coherence frames* $F \in \Sigma$, and *frame-invariant quantities as physical facts*. Collapse, coherence, and classicality are frame-dependent descriptions; Born measures, inner products, spectra, and entanglement entropy are frame-invariant and constitute observer-independent reality.

VII. CONCLUSION

We have introduced coherence relativity, a geometric framework in which:

1. Quantum coherence is frame-relative, like simultaneity in special relativity.
2. The Born rule $\mu_F(i) = |c_i|^2$ is the unique frame-invariant probability measure on the coherence manifold Σ.
3. Wave function collapse is a frame transformation, not a dynamical process.

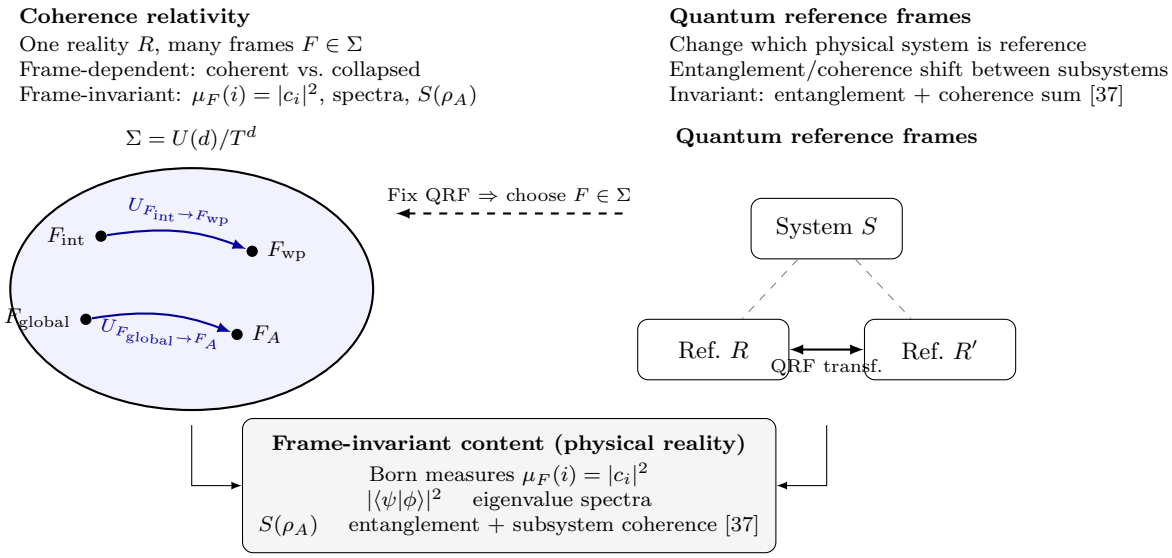

FIG. 6. Coherence relativity vs. quantum reference frames. Left: the coherence manifold Σ with frames F connected by transformations $U_{F\to F'}$. Coherence and collapse are frame-dependent; Born measures and spectra are invariant. Right: QRF transformations change which physical system serves as reference, redistributing entanglement and coherence between subsystems. Dashed arrow: given any QRF choice, one still selects a coherence frame $F \in \Sigma$ on the resulting Hilbert space. Bottom: physical reality consists of quantities invariant under both types of transformation.

4. The information geometry of Σ, encoded in the Fubini–Study metric and the tensor field T_{AB}, generates a quasipotential landscape explaining einselection and classical emergence.

The framework unifies insights from Gleason, Zurek, and operational reconstruction programs while identifying six experimentally testable signatures. The most accessible—frame-transformation hysteresis and measurable transformation timescales—are within reach of current superconducting qubit and trapped ion experiments.

Paper 2 will develop the full tensor field T_{AB} on $\mathcal{M}\times\Sigma$, including the explicit derivation of $T_{M\Sigma}$, the $\delta\lambda$ formalism separating frame-lag from decoherence progress, the coordinate-invariant Markov transition criterion, the mixed-states extension of the Born rule via purification, equations of motion for frame dynamics, and holographic connections between coherence-space geometry and emergent spacetime.

DECLARATIONS

Competing Interests

The author declares no competing financial or non-financial interests.

Data Availability Statement

All numerical data supporting the results in this study (Fubini–Study metric calculations, quasipotential landscapes, and figure generation) are available from the author upon reasonable request. The Python scripts used to generate these data are openly available at `https://github.com/bryankaufman/coherence-relativity`. No experimental datasets were generated or analyzed in this theoretical study.

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Author Contributions

B.K. conceived the theoretical framework, performed all calculations, generated all figures, and wrote the manuscript.

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The conceptual foundations of this work rest on contributions from many giants of quantum theory. Von Neumann's mathematical formalization of quantum mechanics [2] established the measurement problem in its modern form. Bohr's insistence on the role of the observer and Einstein's demand for a complete account of physical reality [24] defined the tension this framework addresses. Bell's penetrating analysis of what quantum mechanics actually requires of any physical theory [1] shaped the standard against which interpretations must be evaluated. Everett's relative-state formulation [30] demonstrated that unitary quantum mechanics alone can accommodate the observer, and Gleason's theorem [3] established the mathematical uniqueness of the Born measure. Zurek's decoherence program [4] and Penrose's gravitational collapse proposal [15] represent the two most developed attempts to locate collapse within physics, and both inform the testable predictions made here. The geometric perspective owes much to the information-geometric program developed by Brody and Hughston [17] and to Bengtsson and Życzkowski's synthesis [16]. The author thanks the members of the Society of Mind coordination network for research assistance and computational support, and is grateful to the open-source scientific Python community for tools enabling the numerical calculations presented herein. A preprint of this manuscript is available at Zenodo: `https://doi.org/10.5281/zenodo.18675342`.

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